

PPOM2

Lecture #80: Thoracic Outlet Syndrome

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Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Session Objectives

- At the end of the session, the student should be able to:
 1. Describe the anatomical landmarks and the 3 anatomical triangles that can potentially cause thoracic outlet syndrome
 2. Deduce the most likely area of entrapment contributing to a patient's symptoms based on information gathered in the history
 3. Diagnose thoracic outlet syndrome based on patient history and physical exam findings
 4. Formulate an osteopathic manipulative treatment plan for a patient with thoracic outlet syndrome based on osteopathic structural considerations

Instruction: *Before* you start this lecture, please read the required article below:

- Attaar N, Pascarella L. Neurogenic Thoracic Outlet Syndrome: A Current Literature Review. Am Surg. 2025 Dec;91(12):2164-2172. Doi: 10.1177/00031348251358432. Epub 2025 Jul 8. PMID: 40626394.

Link: [Neurogenic Thoracic Outlet Syndrome: A Current Literature Review](#)

******This article discusses various tests to diagnose neurogenic thoracic outlet syndrome and various treatments. You are responsible for the tests and management that I cover in the lecture, but just be aware that many options are available. Learn the major points of the article and use the lecture learning objectives as a guide.

*****On pg. 2166 of the article (Section on "Workup", near the bottom of the page): "EEG studies in nTOS patients..." should read "Electrophysiological studies in nTOS patients..."

Recommended resources:

- 1) Jones MR, Prabhakar A, Viswanth O, Uritis I, Green JB, Kendrick JB, Brunk AJ, Eng MR, Orhurhu V, Cornett EM, Kaye AD. Thoracic Outlet Syndrome: a comprehensive review of pathophysiology, diagnosis, and treatment. *Pain Ther.* 2019. 8: 5-18.
- 2) Seffinger MA, Elkiss M. 43 D Osteopathic Considerations in Neurology-Thoracic Outlet Syndrome. In: *Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research.* Philadelphia, PA: Wolters Kluwer; 2018:1278-1279.

Thoracic Outlet Syndrome (TOS)

- Group of diverse conditions with variable symptomology that results from compression of the neurovascular bundle
- What makes thoracic outlet syndrome (TOS) special?
 - A. Uncommon
 - B. Significant symptoms
 - C. Diagnostic challenge
 - D. Illustrates importance of anatomical knowledge and reasoning

College of Osteopathic
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Symptoms of TOS include the following:

- Pain in shoulder and/or upper extremity
 - Paresthesia (e.g. numbness, tingling, burning); Coldness
 - Pallor
 - Provoked by certain motions
 - Poor function (weakness, heaviness, fatigue, lack of endurance, atrophy)
 - Particular side
-
- Any general thoughts? (many sx's but somehow tied together, many conditions can cause this!)

Types of TOS

- Neurogenic
- Vascular
 - Venous
 - Arterial

Types of TOS

- **Neurogenic:** compression of the brachial plexus (C5-T1)
 - Most often lower plexus (C8 and T1)- severe weakness and muscle atrophy may result if chronic
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 - Venous
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What is the *most common* type of TOS and its *percentage of cases*?

Types of TOS

- **Neurogenic:** compression of the brachial plexus (C5-T1)
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What is the *most common* type of TOS (Neurogenic) and its *percentage of cases* (>90%)?

History taking: being the detective

Incidence of thoracic outlet syndrome (TOS):

- A. NTOS: 2-3 cases per 100,000 people/yr
- B. VTOS: 0.5-1 case per 100,000 people/yr
- C. ATOS: Sporadic



Illig KA et al. The Incidence of Thoracic Outlet Syndrome. Ann Vasc Surg. 2021 Jan;70:263-272.

History taking: being the detective

Epidemiology of thoracic outlet syndrome (TOS):

- A. Typical onset in those 20-50 years old
- B. Female > Male



History taking: being the detective

Classification of thoracic outlet syndrome (TOS):

- A. Congenital
- B. Acquired
 - Traumatic
 - Functional



History taking: being the detective



What should we look for on the history?

- A. Types and distribution of symptoms (including laterality)
- B. Provoking factors (movement induced typically)
- C. Timing of symptoms
- D. Daily activities; any change *before and since* symptom onset?
- E. Trauma

History taking: being the detective

What *else* should we look for on the history (or be mindful of)?

- A. Smoking status
- B. Advanced age
- C. Severe, rapidly progressive symptoms
- D. Non-musculoskeletal symptoms
- E. Constitutional symptoms and red flags (eg. weight loss, night sweats, dyspnea, hemoptysis)



Anatomical considerations in TOS: What's in a name?

Consider other names that have been used to refer to TOS:

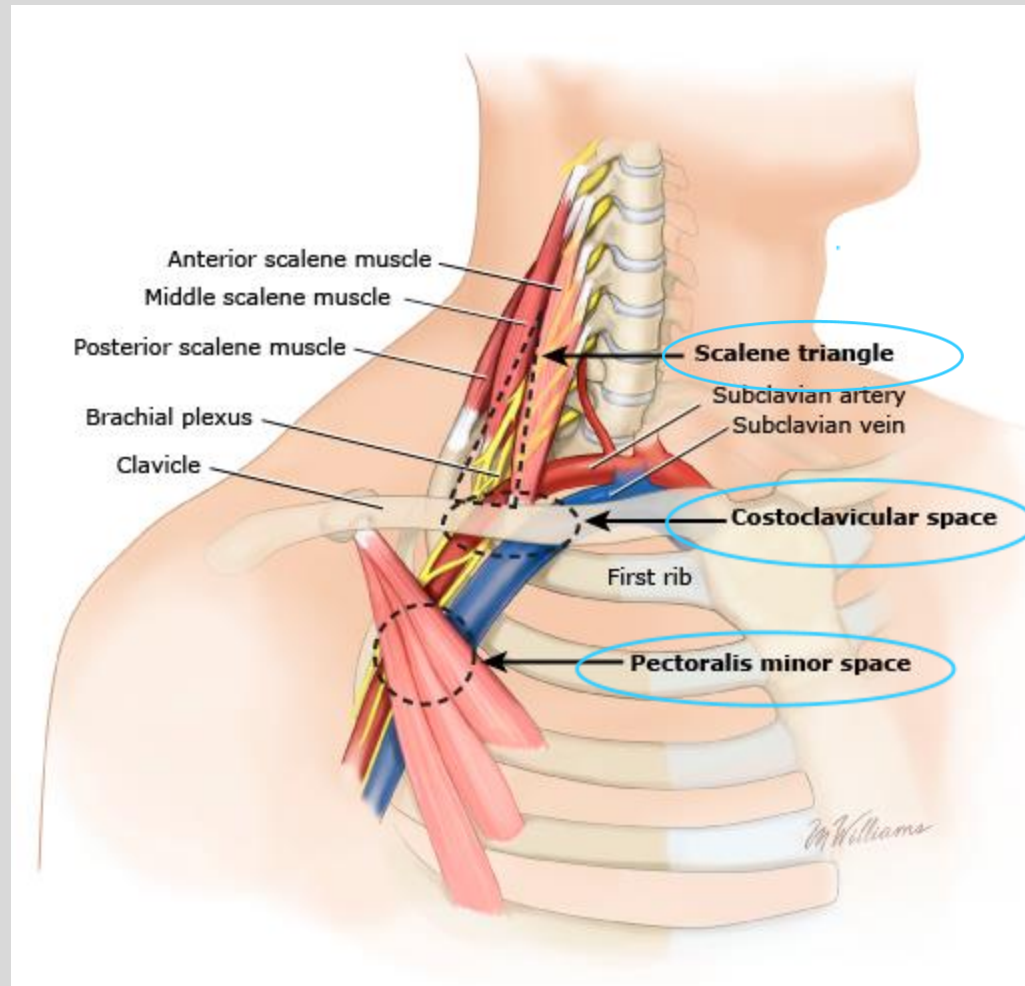
- Cervical rib syndrome
- Scalenus anticus syndrome
- Costoclavicular space syndrome
- First rib syndrome
- Paget-von Schroetter syndrome (aka venous TOS, primary effort thrombosis, etc.)

Anatomical considerations in TOS:

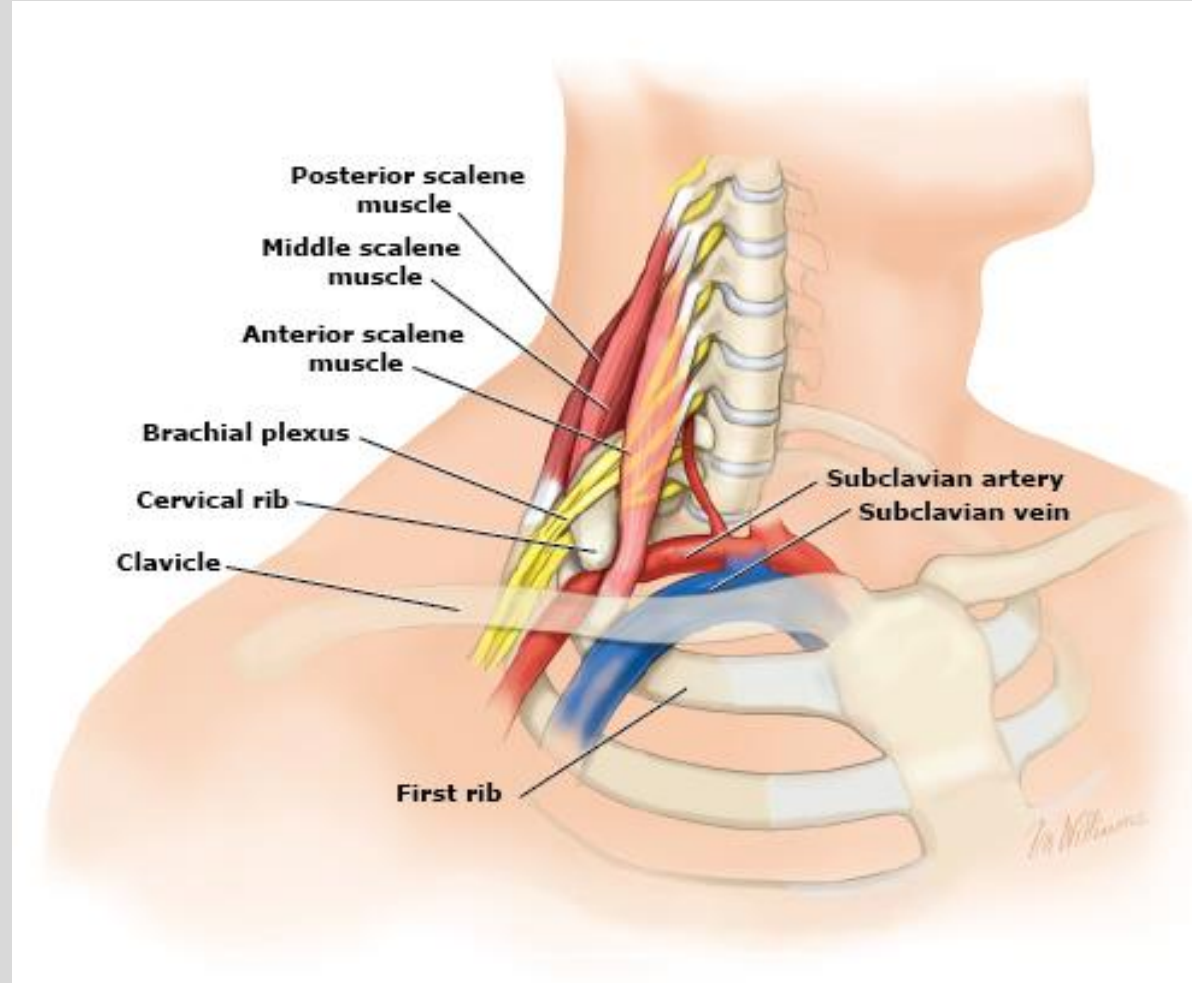
Three primary regions of entrapment

- Interscalene triangle (aka scalene triangle)
 - Brachial plexus and Subclavian artery pass through
 - Most common site of **brachial plexus impingement**
- Costoclavicular space
 - Brachial plexus, Subclavian artery and Subclavian vein pass through
 - Most commonly compressed at this site is **subclavian vein**
- Subcoracoid space (aka retropectoralis space or subcoracoid pectoralis minor space)
 - Brachial plexus, Axillary artery and Axillary vein pass through
 - **Neurovascular bundle compression** almost as common as in interscalene triangle

Three primary regions of entrapment in TOS



Cervical rib causing TOS



Symptoms/signs and triggers for Neurogenic TOS

- **About 90% of total cases of TOS; *most often* from compression of the brachial plexus in the anterior scalene triangle**

- Upper extremity paresthesia (98%)
- Neck pain (88%)
- Trapezius pain (92%)
- Shoulder and/or arm pain (88%)
- Supraclavicular pain (76%)
- Chest pain (72%)
- Occipital headache (76%)
- Paresthesia in fingers (58% in all; 26% in 4th and 5th only)
- Muscle atrophy
- Scalene tenderness

- Repetitive activity (esp. if vigorous or overhead)
- Heavy lifting
- Prolonged computer work
- Driving

Symptoms/signs and triggers for Venous TOS (aka Paget-von Schroetter syndrome or “Effort Thrombosis”)

- **10-15% of cases; from compression of the subclavian vein within the costoclavicular space**

- Forearm fatigue within minutes of using arm may be present
- UE **swelling**, cyanosis, heaviness
- UE paresthesia and/or pain
- Swelling in shoulder, supraclavicular, or chest may be present; venous patterning if vein stenosis/occlusion
- Consider thrombosis

- Repetitive activity (esp. if vigorous or overhead)
- Heavy lifting
- Spontaneous vs trauma-induced
- Extrinsic compression (eg. tumor)

Symptoms/signs and triggers for Arterial TOS

- **2-5% of cases; from compression of the subclavian artery within the scalene triangle**

- Pallor
- Pain
- Paresthesia
- Coldness
- Decreased BP
- Diminished or absent pulse
- Sxs not elicited by provocative maneuvers
- Scalene not tender
- Consider aneurysm (eg. aneurismal pulse) or thrombus

- Almost always associated with a cervical rib or anomalous rib
- Not work or trauma related
- However, vigorous activity may be a factor in some (eg. athletes)

Based on the historical and anatomical considerations of TOS, what should the immediate focus of your clinical evaluation be?

Based on the clinical historical and anatomical considerations of TOS, what should the immediate focus of your clinical evaluation be?

- A. Identify potentially serious or limb threatening conditions
- B. Neurological status
- C. Vascular status

Differential diagnosis of TOS

- Cervical disorder
- Shoulder disorder
- Upper extremity disorder
- Upper thoracic disorder
- Organ disorder (eg. referred pain from visceral cause)

Differential diagnosis of TOS-

Visceral causes include:

- Cardiac
- GI
- Pulmonary
- Reproductive

Adson's test: What does it test?

- Purpose:

- “to determine the state of the subclavian artery, which may be compressed by an extra cervical rib or by tightened anterior and middle scalene muscles” (*FOM, 4th ed: Osteopathic considerations in Neurology*)
- For “brachial plexus impingement between (the) anterior and middle scalene” (*The 5-Minute Osteopathic Manipulative Medicine Consult, 2e > Specialized Tests: Head, Cervical, Lumbar, Lower Extremity, Innominate, Sacrum and Upper Extremity*)

Adson's test:

- Stand behind the patient
- Stabilize the patient's shoulder and clavicle
- Extend the patient's shoulder while monitoring the radial pulse
- **Positive test:**
The radial pulse significantly decreases or is absent, or if the test exacerbates the patient's pain or paresthesia



The 5-Minute Osteopathic Manipulative Medicine Consult, 2e:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=216698488&bookid=2651#216698650>

Table 2 Common provocative diagnostic tests for thoracic outlet syndrome

Test	Maneuver	Result
ADSON TEST	Affected arm is abducted 30° at the shoulder while maximally extended. While extending the neck and turning head towards ipsilateral shoulder, patient inhales deeply	Decrease or absence of ipsilateral radial pulse
Elevated Arm Stress Test (EAST) or ROOS	Arms are placed in the surrender position with shoulders abducted to 90° and in external rotation, with elbows flexed to 90°. Patient slowly opens and closes hand for 3 min	Precipitates pain, paresthesias, heaviness or weakness
Upper Limb Tension Test (ULTT) or ELVEY	Position 1: arms abducted to 90° with elbows flexed Position 2: active dorsiflexion of both wrists Position 3: head is tilted ear to shoulder, in both directions	Positions 1 and 2 elicit symptoms on the ipsilateral side, while position 3 years elicits symptoms on the contralateral side

Jones MR et. al. Thoracic Outlet Syndrome: a comprehensive review of pathophysiology, diagnosis, and treatment. *Pain Ther.* 2019

Adson's test:

- False positive
- False negative
- Remedies include:
 - ✓ Take a good history and do a good physical exam!
 - ✓ Do a combination of tests
 - Eg. Spurling's test, Tinel's sign (aka Hoffman-Tinel sign)

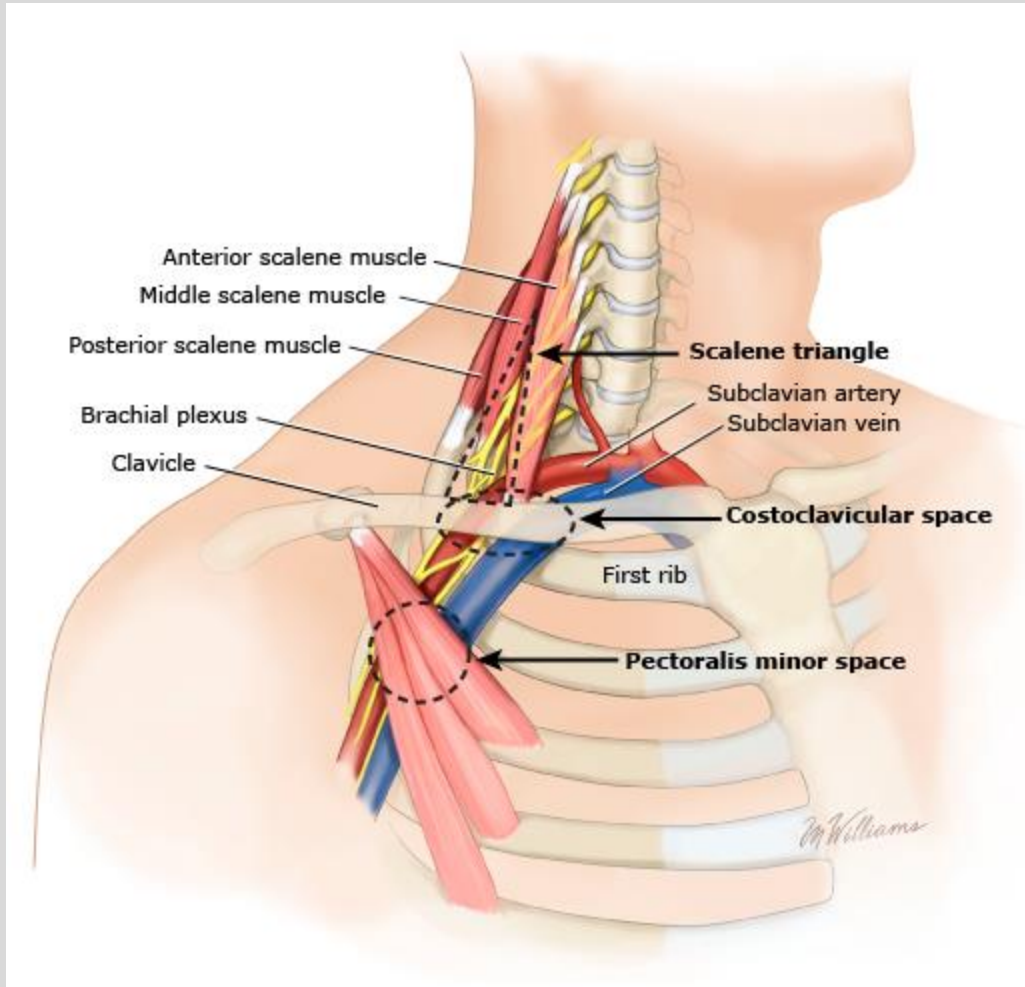
Diagnostic methods for TOS

- Guided by your history and physical exam
- Methods include:
 - Radiographs: spine, chest, shoulder, upper extremity
 - MR, CT, MRN
 - Ultrasound
 - Electrophysiological studies (EMG/NCS)
 - Arteriography, angiography
 - Hemodynamic testing
 - Local anesthetic injection into anterior scalene
 - 94% positive outcome after surgery in those with positive response to block (vs. 50% response in those who failed block)

Diagnostic goals

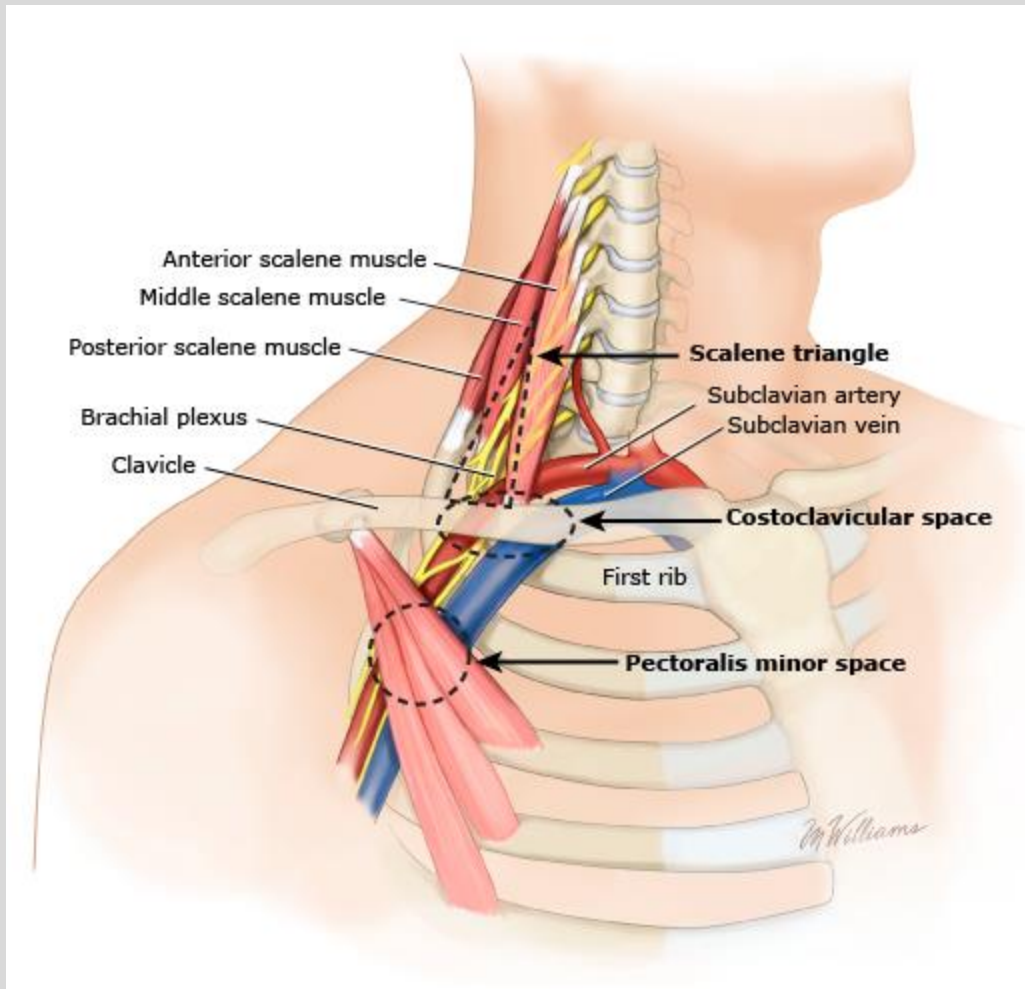
- Determine if alternative diagnosis can better explain the patient's symptoms (most, if not all)
 - Think proximal and distal compressive neuropathies (Spurling maneuver, Tinel's sign, Phalen's sign) and other causes (eg. cervical arthritis) that may cause sx's
 - Think of somatic dysfunctions that can impede anatomical relationships and anatomical functioning
- Determine if an alternative diagnosis is *coincidental (concomitant)* or *causative* to the patient's symptoms/signs
- We can then treat the condition by addressing the cause(s), and improving symptoms and function

Revisiting: The 3 primary regions of entrapment in TOS



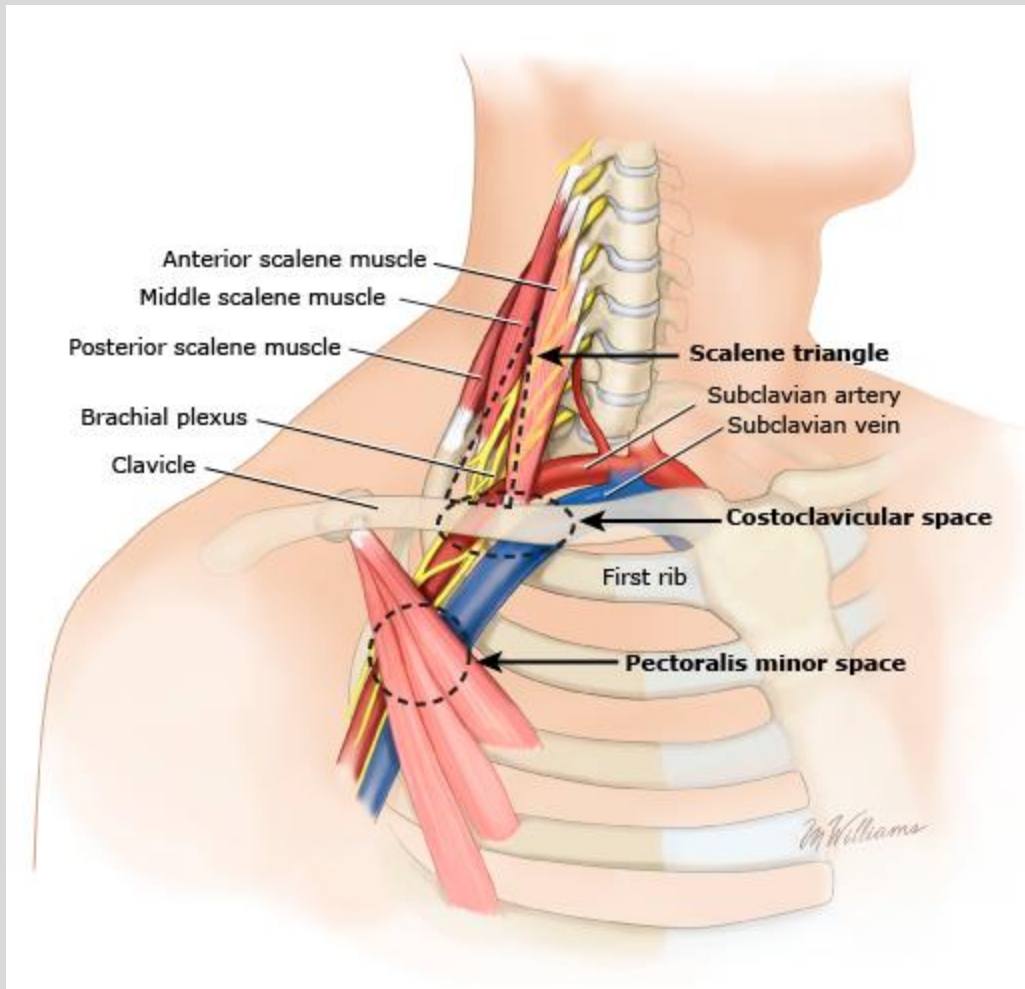
- Interscalene triangle (aka scalene triangle):
 - Anterior scalene
 - Middle scalene
 - First rib (ie. rib 1)

Revisiting: The 3 primary regions of entrapment in TOS



- Costoclavicular space:
 - Subclavius (anterior border)
 - Clavicle (superior border)
 - Anterior scalene (posterior border)
 - First rib (inferior border)

Revisiting: The 3 primary regions of entrapment in TOS



- Subcoracoid space:
 - Coracoid (superior border)
 - Pectoralis minor (anterior border)
 - Ribs (posterior border)

Self Assessment:

- What is the most common type of TOS?
- What is the most common site of entrapment in TOS?
- What are the structures you should most consider in TOS?

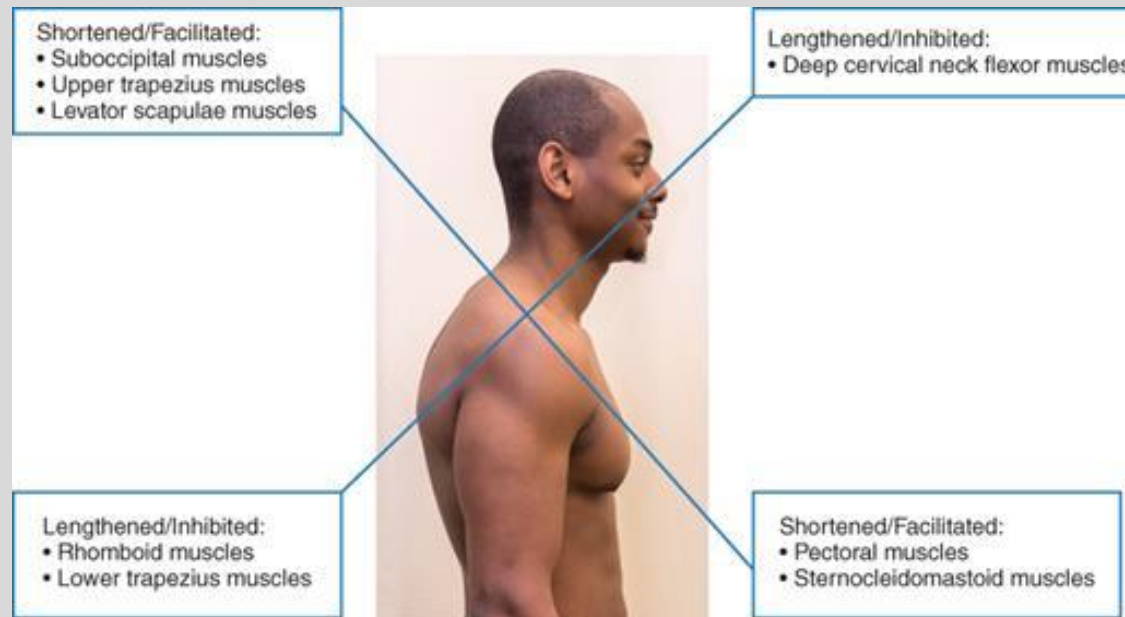
Answers:

- What is the most common type of TOS? *Neurogenic*
- What is the most common site of entrapment in TOS? *Scalene triangle; Subcoracoid space*
- What are the structures you should most consider in TOS? *Anterior and middle scalenes, rib 1, pectoralis minor*

Other important structures to consider:

- Posture
- Cervical spine (C3-6 for anterior scalenes; C2-7 for middle scalenes)
- Clavicle
- Thoracic spine (sympathetic innerv. to UEs; impact on ribs)
- Upper extremity
 - Clavicle
 - Shoulder, Scapula
 - Elbow, wrist, hand
 - Soft tissue findings (eg. changes in color, temp, hair pattern, texture, atrophy, etc.)

The Osteopathic Approach: Bridging what you see with what you know (about form and function)



Travell, Simons & Simons' Myofascial Pain and Dysfunction: The Trigger Point Manual, 3e > Chp. 76 Postural Considerations: "The Living Engine Light"

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=250177415&bookid=3060#250177571>

Department of Osteopathic Manipulative Medicine

Management:

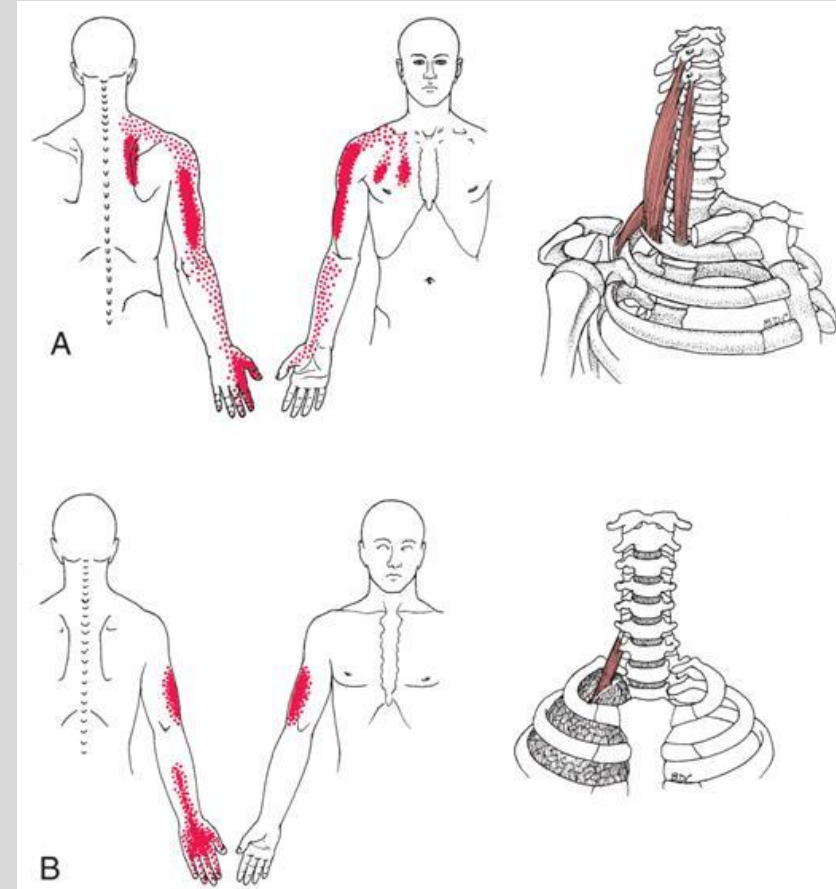
- Conservative
 - Physical rehabilitation (posture, stretching, strengthening)
 - OMT
 - Activity modification, ergonomics
 - Stress reduction
 - Weight control
 - Pharmacological (eg. NSAIDs, muscle relaxants, antidepressants, anticonvulsants; *thrombolysis, heparin*)
 - Injection (analgesics, steroid, including botulinum toxin type A)
 - Patient education and monitoring are key
 - Adequate trial of 4-6 months (usually)

Management:

- Surgical (vascular surgeon typically)
 - Failed conservative management
 - Arterial or Venous TOS- often initial management is surgical
 - Neurogenic TOS- rib 1 resection (often with scalenectomy or scalenotomy)
 - Pros: “excellent” results in 95% of cases in nTOS; >95% patency rate in vTOS 5 years after surgery
 - Cons: complications can occur (eg. brachial plexus injury)

An osteopathic approach- continued

- Recognize possible pain referral patterns
- Learn to palpate and identify anatomy with precision



Travell, Simons & Simons' Myofascial Pain and Dysfunction: The Trigger Point Manual, 3e>
Chp. 20, Scalene Muscles: "Pseudo-Radiculopathy"

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=250171517&bookid=3060#250171547>

An osteopathic approach-continued

- Techniques that can be used include
 - Myofascial release (direct or indirect)
 - Soft tissue
 - Balanced ligamentous tension
 - Counterstrain
 - HVLA

In all cases, think through indications and contraindications, and adapt the technique to the patient

Cervical myofascial release

Findings may include:

- Cervical segmental SD (eg. C4 FRSleft)
- Cervical muscle hypertonicity (eg. paraspinal, scalene)



Beatty, DR et al, The Pocket Manual of OMT, 2e > Chp. 9 Cervical Diagnosis and Treatment

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=49589358&bookid=814#49590681>

Thoracic inlet release



Greenman's Principles of Manual Medicine, 5e> Chp. 11 Principles of Myofascial Release and Integrated Neuromusculoskeletal Technique:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=116099096&bookid=1743#116099147>

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Somatic Dysfunction in Osteopathic Family Medicine, 2e> Chp. 19 The Hospitalized Patient:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=61529168&bookid=1015#nelson2-ch019-fig003>

Rib 1 inhalation MET

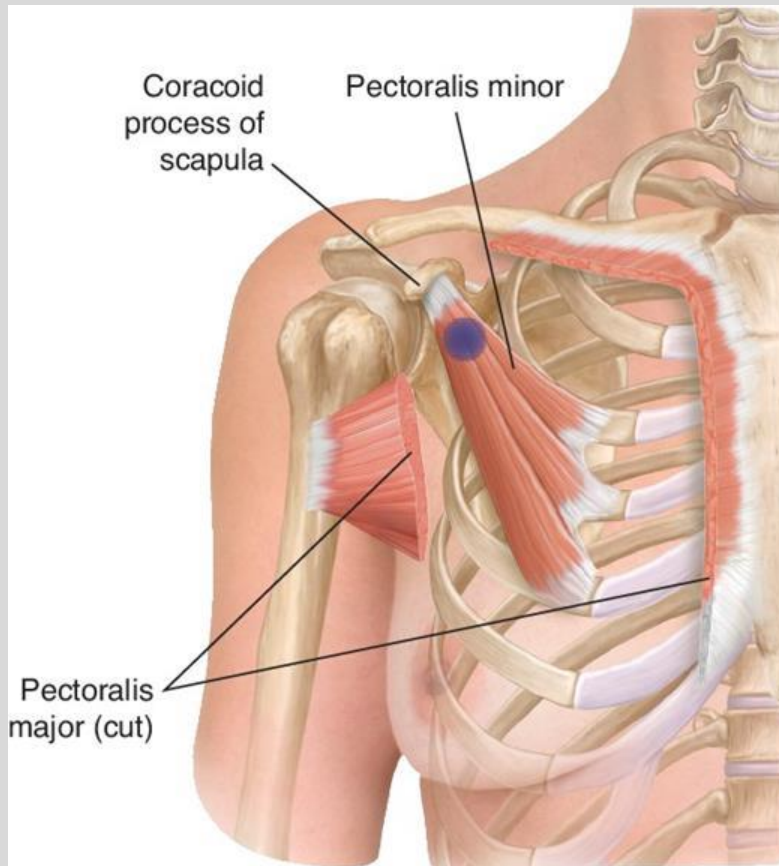


Nicholas A, Nicholas E, Atlas of Osteopathic Techniques, 4th edition, Wolters Kluwer 2023: Chapter 10:

Muscle Energy Techniques LINK:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=252634405&bookid=3202#252634543>

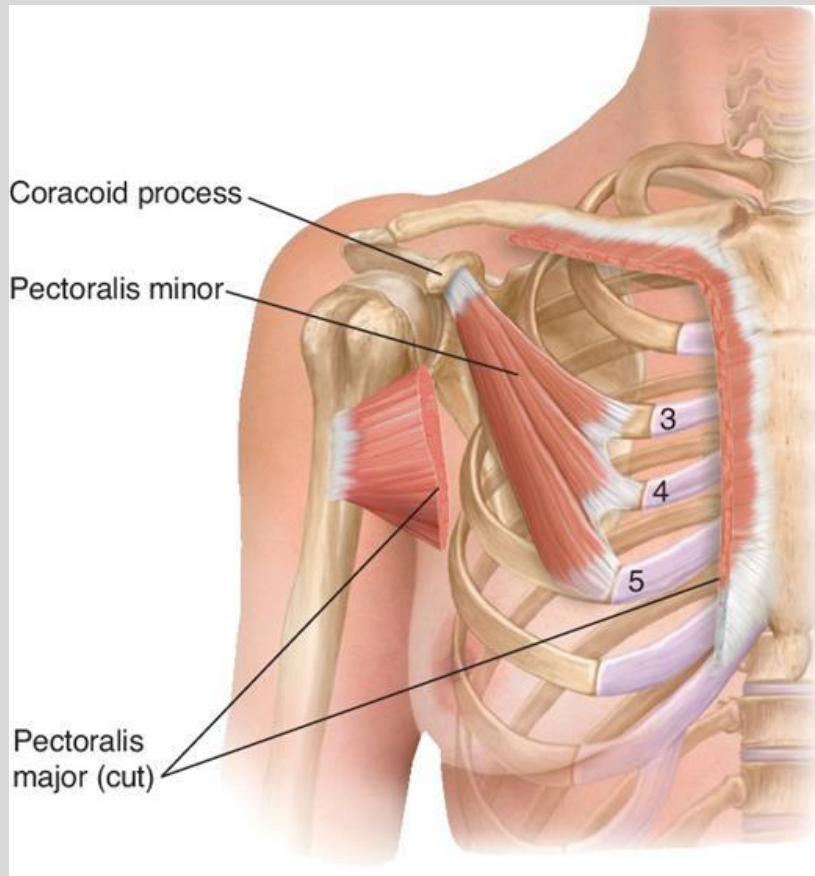
Counterstrain for Pectoralis Minor



Nicholas A and Nicholas E. *Atlas of Osteopathic Techniques*, 4ed edition, Chp. 9-Counterstrain Techniques

<https://login.nyit.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/content.aspx?sectionid=252633080&bookid=3202&resultClick=1#252634122>

MET for the Pectoralis Minor



Nicholas A and Nicholas E. *Atlas of Osteopathic Techniques*, 4ed edition, Chp. 10- Muscle Energy Techniques

<https://login.nyit.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/content.aspx?bookId=3202§ionId=252634405&resultClick=1#252634543>

After completing treatment

- Reassess the SD
- Reassess (over time) patient response
 - ✓ Symptoms
 - ✓ Signs
 - ✓ Function

Summary

1. Thoracic outlet syndrome is an uncommon but important diagnosis to consider in patients with appropriate symptoms, especially if they are severe, worsen, or become chronic
2. It affects both the young (eg. teenagers) and old (50-60). In the young to middle aged, ask about vigorous or repetitive upper body activities, especially involving motions above shoulder level.
3. Diagnosis is challenging, but knowing the prevalence of the various types of TOS, and performing a detailed history and physical exam, including the osteopathic structural exam, can help you to determine the cause and aid in its management
4. Given the physical causes of thoracic outlet syndrome, the osteopathic manipulative approach of using look-feel-move can (and should) be incorporated into your evaluation to enhance your ability to diagnose and treat modifiable factors in the patient's care.

References

1. Assigned reading as previously indicated
2. Goshima Kaoru, Overview of Thoracic Outlet Syndromes, UpToDate, last updated Jan 2025
3. Illig KA, Rodriguez-Zoppi E, Bland T, Muftah M, Jospitre E. The Incidence of Thoracic Outlet Syndrome. *Ann Vasc Surg.* 2021 Jan;70:263-272. doi: 10/1016/j.avsg.2020.07.029. Epub 2020 Aug 6. PMID: 32771464.
4. Jones MR, Prabhakar A, Viswanth O, Uritis I, Green JB, Kendrick JB, Brunk AJ, Eng MR, Orhurhu V, CornettEM, Kaye AD. Thoracic Outlet Syndrome: a comprehensive review of pathophysiology, diagnosis, and treatment. *Pain Ther.* 2019. 8: 5-18.
5. Joseph M. Donnelly, César Fernández de las Peñas, Michelle Finnegan, Jennifer L. Freeman, Travell, Simons & Simons' Myofascial Pain and Dysfunction: The Trigger Point Manual, 3e, Wolters & Kluwer 2019

Thank you

Lecture and Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>